Day-21 -Regular Expression

📌 1. Introduction to Regular Expressions

🔹 What is Regex?

Regular Expressions (Regex) are used to **search, match, and manipulate text patterns**.

👉 In simple words:

> Regex helps you **find specific patterns in text automatically**

🔹 Why Use Regex?

- Search text quickly
- Validate data (email, phone, password)
- Extract useful information
- Replace or clean text

👉 Example:

- Find all emails in a document
- Check if password is strong

📌 2. Regex in Python

Python provides a built-in module:

```
importre
```

👉 This module allows you to use regex functions easily.

📌 3. Important Regex Functions

🔹 1. `re.match()`

- Matches pattern **only at the beginning**

```
importre

result=re.match("Hello","Hello World")
print(result)
```

✅ Match → because string starts with "Hello"

🔹 2. `re.search()`

- Searches pattern **anywhere in string**

```
importre

result=re.search("Hello","Say Hello")
print(result)
```

✅ Match → found inside string

🔹 3. `re.findall()`

- Returns **all matches as a list**

```
importre

result=re.findall(r"\d","abc123")
print(result)
```

👉 Output:

```
['1', '2', '3']
```

📌 4. Match vs Search (Important Difference ⚠️)

Function	Behavior
`match()`	Checks only at start
`search()`	Checks entire string

👉 This is a very common exam & interview question.

📌 5. Common Regex Symbols

Symbol	Meaning
`^`	Start of string
`$`	End of string
`*`	0 or more times
`+`	1 or more times
`.`	Any character

🔹 Example

```
r"^Hello"
```

👉 String must start with "Hello"

📌 6. Special Sequences

Symbol	Meaning
`\d`	Digit (0–9)
`\s`	Whitespace
`\w`	Word (letters + digits + _)

🔹 Example

```
r"\d"
```

👉 Matches any number

📌 7. Character Sets

🔹 Definition

Used to match specific characters or ranges.

🔹 Example

```
r"[a-z]"
```

👉 Matches all lowercase letters

```
r"[0-9]"
```

👉 Matches digits

📌 8. Regex Flags

Flags change how regex works.

🔹 Example: Ignore Case

```
importre

result=re.search("hello","HELLO",re.IGNORECASE)
print(result)
```

👉 Matches even with different cases

📌 9. Real-Life Applications

✅ 1. Email Validation

Check if email format is correct

✅ 2. Password Checker

- Length check
- Uppercase, lowercase, numbers

✅ 3. Product Search Filters

Used in:

- Amazon
- Flipkart
- Search engines

📌 10. Practical Examples

🔹 Find Numbers

```
importre

text="My number is 987654"

numbers=re.findall(r"\d",text)
print(numbers)
```

🔹 Email Validation

```
importre

email="test@gmail.com"

pattern=r"^[\w\.-]+@[\w\.-]+\.\w+$"

ifre.match(pattern,email):
print("Valid Email")
```

```
else:
print("Invalid Email")
```

📌 11. Common Mistakes ⚠️

- Using `match()` instead of `search()`
- Forgetting raw string (`r""`)
- Wrong pattern syntax
- Not testing regex properly

📌 12. Mini Practice Activity

🧠 Try Yourself:

1. Find all numbers in:

   ```
   "Order 123, Price 456"
   ```

2. Validate emails from list:

   ```
   test@gmail.com
   test@.com
   user123@yahoo.in
   ```

📌 13. Final Understanding

If a student understands:

- Patterns
- Symbols
- Functions

👉 They can:

- Validate data
- Build smart applications

- Work with real-world text data

📌 14. Key Takeaway

> Regex is not just coding — it is **pattern thinking**

And this skill is used in:

- Backend development
- Data science
- Cybersecurity
- Automation